

3.5 hours

Problem 1 (20TSTST1)

We claim the answer is $\max(ab, bc, ca)$, attained by seating the ducks facing clockwise, and, in clockwise order, placing a block of all ducks picking rock followed by a block of all ducks picking scissors following by a block of all ducks picking paper. If WLOG $ab = \max(ab, bc, ca)$ for example, we may achieve the maximum by moving the counter-clockwise-most duck picking scissor through all the a ducks picking rock. Doing this b times over all ducks picking scissor, we finish.

WLOG, let the ducks each be facing clockwise. We count the number of triplets t of ducks (x, y, z) where x picks rock, y picks scissor, and z picks paper where x, y, z are seated clockwise in that order. The maximum possible value of t is abc since that's the number of triplets of ducks picking rock, scissor, or paper respectively ignoring the clockwise condition.

We see that every time we switch a pair of ducks one of rock and one with scissors, all triplets involving those two ducks and one of the c ducks picking paper must have previously been in clockwise order. After the swap, they are not. All other triplets are unaffected by the swap since we are just swapping the position of those two ducks alone. Therefore, t decrements by c .

If we instead switch pairs of ducks one with rock and one with paper, we decrement by b .

If we switch pairs scissor and paper, we decrement by a .

Clearly, to maximum number of moves, we decrement by $\min(a, b, c)$ so the upper-bound is $\max(ab, bc, ca)$ as desired since $t \leq abc$.

Problem 2 (00AMO3)

We denote a sequence of moves as a string such as $wrrrrrbbbwwww$ where each character represents the card played. r, w, b stands for red, white, blue cards respectively and taking power x means repeating the character x times. The penalty is the number of pairs b, w where the w occurs after the b plus two times the number of pairs wr in that order plus three times the number of pairs rb in that order.

We claim that the minimum is $\min(BW, 2RW, 3BR)$, attained by taking one of the strings $r^R w^W b^B, w^W b^B r^R, b^B r^R w^W$.

We claim the number of ways in which this can be attained is,

- One if $\min(BW, 2RW, 3BR)$ is unique, with the way to attain already shown
- $W + 1$ if $BW = 2RW$, with taking $w^x b^B r^R w^{W-x}$ for any $0 \leq x \leq w$ work
- $B + 1$ if $BW = 3BR$ with the analogous constructions
- $R + 1$ if $3BR = 2RW$ with the analogous constructions
- $W + B + R$ if $BW = 2RW = 3BR$ with the union of the above constructions

We now prove that this is the minimum and no other minimums are possible.

Lemma: No consecutive “ wr ”, “ rb ”, or “ bw ” may be present in a construction for the minimum.

Proof: If they are present, simply swap the two characters to reduce the sum by 2, 3, 1 respectively as we'll take away one of the penalty pairs without affecting anything else.

We find that the minimum string must be one of, for $x_1, x_2, \dots > 0$ and some integer n ,

- $r^{x_1} w^{x_2} b^{x_3} r^{x_4} \dots$
- $w^{x_1} b^{x_2} r^{x_3} w^{x_4} \dots$
- $b^{x_1} r^{x_2} w^{x_3} b^{x_4} \dots$

Call continuous substrings of consecutive characters bounded on each side by different characters or nothing such as r^{x_1}, w^{x_2} blocks. $\square$

Claim:

- i. There exists a string achieving the minimum with at most 3 blocks.
- ii. There exists no string with at least 5 blocks achieving the minimum.
- iii. There only exists strings of 4 blocks achieving the minimum if it is of the $r^{x_1} w^{x_2} b^{x_3} r^{x_4}$ form and $2RW = 3BR$ or analogous forms starting with w, b as mentioned when we listed the number of ways in each case.

Proof: Consider WLOG (we will revisit later why we can do this), a string $r^{x_1} w^{x_2} b^{x_3} r^{x_4} w^{x_5} \dots$. Note that we may instead make the string either,

$$r^{x_1+x_4} w^{x_2} b^{x_3} w^{x_5} \dots \text{ or } w^{x_2} b^{x_3} r^{x_1+x_4} w^{x_5} \dots$$

The former changes the penalty by $x_4(3x_3 - 2x_2)$ while the latter changes it by $x_4(2x_2 - 3x_3)$. Note that we may take WLOG since starting with a different character would just change the coefficients of x_2, x_3 . One of these would be non-negative. Therefore, while there are at least 4 blocks, we may reduce the total number of blocks by one without increasing the penalty. This proves (i).

We now consider (ii). When we start with at least five blocks, for there to be no string with no penalty lower, we must have $x_4(3x_3 - 2x_2) = x_4(2x_2 - 3x_3) = 0$. This means both $r^{x_1+x_4} w^{x_2} b^{x_3} w^{x_5} \dots$ and $w^{x_2} b^{x_3} r^{x_1+x_4} w^{x_5} \dots$ have the same penalty as the string we started with. However, we see that $r^{x_1+x_4} w^{x_2} b^{x_3} w^{x_5}$ can be improved upon further since there is a consecutive pair bw .

For (iii), we see that, with four blocks, if $3x_3 - 2x_2 \neq 0$, one of $x_4(3x_3 - 2x_2)$ or $x_4(2x_2 - 3x_3)$ may be negative so there's another string with lower penalty. Therefore, $3x_3 - 2x_2 = 0$ for the construction to be possible. There are no other blocks involving w, b so $x_3 = B; w = x_2$.

$$3x_3 - 2x_2 = 0 \Leftrightarrow 2W = 3B \Leftrightarrow 2RW = 3BR \quad \square$$

Using the latest claim, we find that nothing other than the constructions we gave are unique.

4 clubs; 5 hours

Problem 3 (15USATST3)

We claim the answer is no. Assume FTSOC there is a sequence of moves and a selection of electrons which achieves this. Where A is the subset of the 2015 atoms that has the electrons, let f be the

function such that for any A , $f(A)$ is the subset of atoms which have electrons at the end if we apply the sequence of moves using the diode on A .

Clearly, since electrons are never destroyed, created, transferred from, or transferred into the system, $|A| = |f(A)|$.

Lemma: If $A \subset B$, then $f(A) \subset f(B)$

Proof: We induct on the number of steps. With zero steps, this trivially holds. With every new step, when an electron in both A and B is moved out of B , then it will be transferred to the same new position in both cases from $A \subset B$. When an electron in A is moved out of A to another position not in B , it is also transferred to the new position in both cases. No other operations may cause $A \subset B$ to turn false so we are done. $\square$

WLOG, let's say our selection of electrons was $\{1, 2\}$. Therefore, for all A , we must have,

$$\{1, 2\} \cap f(A) \in \{\emptyset, \{1, 2\}\}$$

However, starting with $A = \emptyset$ in which case $f(A) = \emptyset$, we add new elements to A one-by-one. Since $|A| = |f(A)|$ and using the lemma, we will be adding new elements to $f(A)$ as well one-by-one. We finish at $A = \{1, \dots, 2015\}$ in which case $f(A) = \{1, \dots, 2015\}$. At some point along the way, we must have added 1 before 2 or 2 before 1 in which case we've found an A that does not satisfy $\{1, 2\} \cap f(A) \in \{\emptyset, \{1, 2\}\}$.

Problem 5 (04SLC5)

The answer is all n such that when n is written in base 4, each digit is either 0 or 2.

Define $f(k, n)$ as a function returning whether the person who's turn it is when the number is k and the goal is n has a strategy to eventually win. We know $f(n, n) = 0$ and we want to find when $f(1, n) = 1$.

Each player's strategy at $f(k, n)$ would be to look at $f(k + 1, n)$ and $f(2k, n)$. They'd try to go for the option that is equal to zero since, for example, if $f(k + 1, n) = 1$, their opponent would eventually win at that state. Instead, if, for example, $f(k + 1, n) = 0$, their opponent cannot win from that state.

Therefore, if either $f(k + 1, n)$ or $f(2k, n) = 0$, they may make the move and eventually force a win. If $f(k + 1, n) = f(2k, n) = 1$ otherwise, then they cannot win in any case.

$$f(k, n) = \text{NOT}(f(k + 1, n) \text{ AND } f(2k, n))$$

For our definition, we define $f(k, n) = 1$ for all $2(n - 1) \geq k > n$ so no player will win through a number to something higher than n .

Lemma: $f(1, n) = 0$ for all odd n .

Proof: We induct backwards from n to see that all $f(k, n) = 0$ for all odd k and $f(k, n) = 1$ for all even k . Since n is odd, $f(n, n) = 0$ clearly works. For even k , $f(k + 1, n) = 0$ meaning $f(k, n) = 1$. For odd k less than n , both $f(2k, n)$ and $f(k + 1, n)$ must be 1 whether $2k > n$ or not so $f(k, n) = 0$. $\square$

Claim: For all even n , $f(1, n) = f\left(1, \frac{n}{4}\right)$ for all $n \equiv 0 \pmod{4}$ and $f(1, n) = f\left(1, \frac{n-2}{4}\right)$ otherwise.

Proof: Start with $f(n, n) = 0$. For all $k > \frac{n}{2}$, we have,

$$\begin{aligned} f(k, n) &= \text{NOT}(f(k+1, n) \text{ AND } f(2k, n)) \\ &= \text{NOT}(f(k+1, n)) \end{aligned}$$

Therefore, $f(n-1, n) = 1$; $f(n-2, n) = 0$ and so on. We have $f\left(\frac{n}{2}, n\right) = 1$ since $f\left(2\left(\frac{n}{2}\right), n\right) = 0$.

If $4 \mid n$,

$$f(n, n) = f(n-2, n) = \dots = f\left(\frac{n}{2} + 2, n\right) = 0$$

Therefore, $f(k, n) = 1$ for all $\frac{n+4}{4} \leq k \leq \frac{n}{2}$ since $f(2k, n) = 0$ in those cases. We'd then get,

$$f\left(\frac{n}{4}, n\right) = \text{NOT}\left(f\left(\frac{n+4}{4}, n\right) \text{ AND } f\left(\frac{n}{2}, n\right)\right) = 0$$

Since $f\left(\frac{n}{4}, n\right) = f\left(\frac{n}{4}, \frac{n}{4}\right) = 0$ and $f(k, n) = f\left(k, \frac{n}{4}\right) = 1$ for all $\frac{n}{4} + 1 \leq k \leq 2\left(\frac{n}{4} - 1\right)$, noting that our transition function $f(k, n) = \text{NOT}(f(k+1, n) \text{ AND } f(2k, n))$ does not actually use the second parameter, we'd get $f(k, n) = f\left(k, \frac{n}{4}\right)$ for all $k < \frac{n}{4}$ as well, including $f(1, n) = f\left(1, \frac{n}{4}\right)$ as desired.

Similarly, if $n \equiv 2 \pmod{4}$ instead,

$$f(n, n) = f(n-2, n) = \dots = f\left(\frac{n}{2} + 1, n\right) = 0$$

We have $f(k, n) = 1$ for all $\frac{n+2}{4} \leq k \leq \frac{n}{2}$ in this case implying $f\left(\frac{n-2}{4}, n\right) = 0$. In this case, we'd get $f(k, n) = f\left(k, \frac{n-2}{4}\right)$ for all $k \leq \frac{n+2}{4}$ for the analogous reason as desired once again. $\square$

Now we see that,

$$f(1, n) = f\left(1, \frac{n}{4}\right) \forall 4 \mid n \Rightarrow f(1, k) = f(1, 4k) \forall k$$

$$f(1, n) = f\left(1, \frac{n-2}{4}\right) \forall n \equiv 2 \pmod{4} \Rightarrow f(1, k) = f(1, 4k+2) \forall k$$

Also,

$$f(1, n) = 0 \forall n \text{ odd} \Rightarrow f(1, 4k+1) = f(1, 4k+3) = 0 \forall k$$

Starting with the base cases $f(1, 1) = 0$; $f(1, 2) = 1$; $f(1, 3) = 0$, we may induct in base 4 to finish.

10 clubs; 8 hours

Problem 7 (16IMO6)

Draw a circle encompassing all the line segments and extend the line segments to be a chord of the circle. We label the $2n$ endpoints of the chord in clockwise order from 1 to $2n$. Each chord goes from

endpoint i to endpoint $i + n$ since it must intersect each of the $n - 1$ other chords. We see if there are ways to place the n frogs at certain points and set them to go to the other side of the chord without colliding.

For part (a), we may select points $1, 3, 5, \dots, 2n - 1$ to place the frogs. For the sake of contradiction, if this does not work, then a frog starting at point i jumping alongside A will intersect one starting at point $i + 2k$ jumping alongside B taking indices modulo $2n$. Let point p be the intersection of A and B . Let a and b be the number of points the frogs will pass through respectively before getting to p . We know $a = b$.

Of the $2k - 1$ points $(i + 1, \dots, i + 2k - 1)$ between the two frogs, if we remove the pairs of them which have chords connecting to each other that do not affect a or b , we have an odd number of points connected to an odd number of distinct chords that connect to points outside of $i + 1, \dots, i + 2k - 1$. Each of these chords may either ((intersect A before its frog reaches p) and (B before its frog reaches p)) or vice versa. Each of these chords hence increment one of a or b by one.

Of the remaining chords which do not interact with $i + 1, \dots, i + 2k - 1$, they either increment both a and b or neither.

We see that $a - b \equiv 1 \pmod{2}$. (!)

For part (b), we claim that we may never place frogs on consecutive points $i, i + 1$. Let a, b once again be the number of points the frogs pass through before intersecting. Any other chord either increments both a, b by one or does not change either. Therefore, the two frogs will collide if make this choice. Therefore, we must either choose $1, 3, \dots, 2n - 1$ or $2, 4, \dots, 2n$. Since n is even, in either way, both i and $i + n$ will be chosen which is not allowed.

Problem 9 (19AM05)

We claim the answer is all pairs (m, n) where $m + n$ is a power of two.

To show that all such pairs work, we see that,

$$\frac{\frac{m}{n} - 1}{\frac{m}{n} - \frac{n}{m}} = \frac{m^2 - mn}{m^2 - n^2} = \frac{m}{m + n}$$

Therefore, since $m + n = 2^k$, we may take the arithmetic average until we are done.

We now show everything else fails. For all other (m, n) , there exists a prime $p \neq 2$ such that $p \mid m + n$. We define the shadow of a number n on the whiteboard to be $n + 1$. When we take the arithmetic mean of two numbers x, y , the shadow of the result $\frac{x+y}{2} + 1 = \frac{(x+1)+(y+1)}{2}$ is the arithmetic mean of the shadows. When we take the harmonic mean of two numbers with shadows x, y , the shadow of the result is $\frac{2(x-1)(y-1)}{x+y-2} + 1 = \frac{2xy-x-}{x+y-2}$. We track the shadows of the numbers on the whiteboard. Initially, we have $\frac{m+n}{n}, \frac{m+n}{m}$ two numbers divisible by p . We want to get 2. We may check that $p \mid x, y \Rightarrow p \mid \frac{2xy-x-}{x+y-2}, \frac{x+y}{2}$ for $p \neq 2$ so it is impossible to attain 2 in the end.

18 clubs; 11.5 hours

Problem 11 (21IMO5)

For the sake of contradiction, let this not be the case. Define the “main character” of the k -th swap be the hole that walnut k is in. We are assuming that the walnut in the main character of each swap either has a lower index than both of its neighbors or higher index. Call a walnut “dead” after its main character moment has happened and “alive” otherwise.

Note that a dead walnut right before the k -th move happens is precisely all walnuts with index less than k while the alive walnuts have index at least k . Using our hypothesis, no swap may be made on an alive and dead walnut. Right after the swap, the main character dies but the statuses of all other walnuts are unchanged. Therefore, if we call a hole dead or alive by the status of the walnut in it, each hole starts alive and only changes once: from alive to dead when the walnut inside it has a main character moment.

When holes die one by one, each of their neighbors must have the same status. We show this is impossible. We use induction to show that it is impossible for this happen to an arc of an even number of alive holes surrounded by dead holes on either side.

Base case: With two alive holes, if either hole has a main character moment, then one of its neighbors is alive and the other is dead so impossible.

Inductive step: Any hole that first dies within the arc must not be the endpoints. For any other choice, it will leave shorter arcs one of odd number of points and one even as desired.

Conclusion: By induction we’re done.

After 1 has its main character moment, we have a chain of 2020 alive holes surrounded by the dead hole on either side. Using the claim, it is impossible to flip each hole to dead.

Problem 14 (19ELMO2)

We claim the answer is all even m such that any prime dividing n divides m . We represent the points as vectors $v_1 = (1, 0, \dots, 0), \dots, v_n = (0, \dots, 0, 1)$.

We first show that no other m work. We take $v_1, \dots, v_n$ such that they are linearly independent over $\mathbb{Q}$. The centroid is $g = \left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right)$.

Step 1: If $p \mid n$ but not m , we see that for any vector we create we cannot get p in the denominator so it is impossible to construct g as desired.

Step 2: For proving m is even, if n is even we may use the previous step. For the sake of contradiction, if m, n are both odd. We may check that all vectors we create are equivalent to one of $v_1, \dots, v_n$ modulo n . Indeed,

$$\frac{kv_i + (m - k)v_j}{m} \equiv kv_i + (1 - k)v_j \equiv v_i \text{ or } v_j \pmod{2}$$

It is impossible to get $g \equiv (1, 1, \dots, 1) \pmod{2}$.

We now show that it is possible to make the construction.

Instead let $v_1 = (p, 0, \dots, 0)$; $v_2 = (0, p, 0, \dots, 0)$; $\dots$; $v_n = (0, 0, \dots, 0, p)$. The centroid is $(1, 1, \dots, 1)$. We start with making $(p - 1, 1, 0, \dots, 0)$ and continue with the following idea:

When there are two different numbers x_i, x_j such that $x_i \equiv x_j \pmod{2}$ but $x_i \neq x_j$ in $(x_1, \dots, x_n)$, we may replace both of them with $\frac{x_i + x_j}{2}$ as follows: by symmetry, we may construct $(x_1, \dots, x_n)$ except with x_i, x_j swapped. Taking the midpoint of the two suffices.

By AM-QM, the sum of squares will decrease with this step. We perform this until we cannot do it anymore. In this case, all the odd numbers have the same value, and all the even numbers have the same value.

Case 1: There is at least one even number. Since the common value of the odd numbers is positive, the common value of the even numbers must be zero for the sum to be n . However, since p is prime, the only way this is possible is $(p, 0, \dots, 0)$ which is impossible to make since $p^2 > (p - 1)^2 + 1^2$ and the sum of squares is decreasing.

Case 2: There are no even numbers. The n odd numbers are all the same and sum to n so we must be at $(1, 1, \dots, 1)$ the desired result.

The only way the process can stop is at the centroid as desired.

28 clubs; 15 hours

Problem 16 (93IMO3)

We claim the answer is all n that are not multiples of threes.

We first show all multiples of three fail. Denote X as a space with a token and 0 as no token. Any move takes $XX0$ to $00X$ or some reflection/rotation of it. We color the grid red, blue, yellow so that any three consecutive cells are different colors. When $3 \mid n$, there is an equal number of red, blue, yellow respectively. We see that the parity of the pairwise differences of the respective numbers is invariant, so it is impossible to get $(1, 0, 0)$ in some order in which two of the differences is odd.

We induct on n for the construction of everything else.

Base case: $n = 1, 2$. $n = 1$ is already done and $n = 2$ can be done as follows:

XX 0XX X

XX $\rightarrow$ X00 $\rightarrow$ X $\rightarrow$ X

Inductive step (i): $n = 3k + 1 \rightarrow n = 3k - 1$ for positive k

Note that we may always empty a streak of three consecutive X lined up if there is another X adjacent to one of the X s not in line with them as follows.

000 00X 00X 000

XXX -> XX0 -> 00X -> 000

00X 000 000 00X

Therefore, given a grid of $3k + 1 \times 3k + 1$ tokens, we may remove the outer border.

XXXXXX

XXXXXX

XXXXXX

XXXXXX

XXXXXX

XXXXXX

XXXXXX

We may remove the red and blue sections one by one to get the $3k - 1 \times 3k - 1$ grid inside. Each one is adjacent to a black X so this is possible.

Inductive step (ii): $n = 3k + 2 \rightarrow n = 3k - 2$ for positive k

Note that we may take,

XXX 000 000

XXX -> XXX -> 000

00X 00X 000

Using this, we remove take the outer border two-wide since each block touches an inner black X which is not removed.

XXXXX

XXXXX

XXXXX

XXXXX

XXXXX

Conclusion: Using induction we are done.

Problem 19 (09SLC5)

We claim that Cinderella may prevent her Stepmother from succeeding with the following strategy. Define a pair of non-consecutive buckets with sum of volume greater of 1 to be “dangerous”. Among the

moves that neutralizes all dangerous pairs by resetting each of their sum back to at most 1, Cinderella picks the one of such moves which clears the maximal amount of water.

We will show that there is a move neutralizing all dangerous pairs of buckets momentarily.

Claim 1: After Cinderella's move, the total sum of the bucket volumes is less than $\frac{3}{2}$.

Proof: For the sake of contradiction, let there be a moment in time when this claim is violated **for the first time**. We call this Cinderella's ultimate move. In other words, prior to this point, the total bucket volume sum has been less than $\frac{3}{2}$ after Cinderella's move implying that the total bucket volume sum after the stepmother's move has been less than $\frac{5}{2}$.

Case 1: Prior to Cinderella's last move, there were no dangerous pairs.

Cinderella simply selects the pair with maximal volume. If the volumes were $a_1 + a_2 + \dots + a_5 < \frac{5}{2}$, by Pigeonhole, $\min(a_1 + a_2 + a_3, a_2 + a_3 + a_4, \dots, a_5 + a_1 + a_2) < \frac{3}{2}$ so it is impossible for this to be the last move.

Case 2: Prior to Cinderella's last move, there was one dangerous pair.

WLOG let the danger be between a_1, a_3 meaning $a_1 + a_3 > 1$. WLOG, let $a_1 \geq a_3$ meaning $a_1 > \frac{1}{2}$. Since there is only one dangerous pair, $a_4 < 1 - a_1$. For this to be the last move, Cinderella has no way of removing at least 1 so assume that $a_1 + a_2 < 1$ meaning $a_2 < \frac{1}{2}$. Since there was only one dangerous pair, $a_1 + a_4 < 1$ meaning $a_4 < 1 - a_1$.

$$\min(a_1 + a_2 + a_3, a_2 + a_3 + a_4, \dots, a_5 + a_1 + a_2) \leq a_4 + a_3 + a_2 < 1 - a_1 + a_3 + \frac{1}{2} \leq \frac{3}{2}$$

... (2 other cases) ...

With the claim proven, we see that is always possible to neutralize all dangerous pairs of buckets.

Clearing a pair of consecutive buckets, say a_1 and a_2 , will neutralize all pairs $a_1, a_3; a_1, a_4; a_2, a_4; a_2, a_5$ except a_3, a_5 . If nothing neutralizes everything, all non-consecutive pairs must sum to > 1 . The overall sum is then $> \frac{3}{2}$.

38 clubs; 18 hours

Problem 21 (19SLC4)

We claim the only possible is $n + 1$. In other words, for any labyrinth, there is a way for Lucy to color the walls such that at most $n + 1$ minions may be placed.

If we let the $\binom{n+1}{2} + 1$ regions we make be vertices on a graph and draw $\binom{n}{2}$ edges between two vertices connected by a door, the number of minions we may place at most is the number of connected

regions there are. There are at least $\binom{n+1}{2} + 1 - \binom{n}{2} = n + 1$ connected components so the lowerbound is complete.

For how Lucy may attain $n + 1$, we draw a circle around all the intersections. Select a “special” region touching the edge of the circle and color all walls such that the region is on the blue side of the edge (do this for all the walls; regardless of whether the wall borders the region or not). For each region, we give it a label which is the number of walls (still doing this for all the walls) which the region is on the red side of the wall. In particular, the special region is labeled zero.

There is a door between two regions if and only if the regions are touching at an intersection and have the same label.

We call the k th line the segments of walls which forms the border between a region labeled k and one labeled $k + 1$ and the arc of the circle that borders a region labeled k . The 0th line is all the walls and arcs of the circle that surround the special region. The regions on the other side of these walls are labeled 1. The 1st line is the wall segments and circle arcs that surround the regions labeled 1 that do not form the 0th line. The regions on the other side of the 1st line are labeled 2 and so on and so forth.

We may induct to see that each k th line is a series of line segments from one point on the circle to another without intersecting itself. Therefore, it is impossible for there to be a cycle of regions with the same label as desired. [clarification needed?]

Problem 23 (20MEX3)

We show that Beto wins if and only if n is prime.

Case 1: n is even.

Since $n \geq 3$, n is not a prime meaning we show that Ana wins. Note that however Ana colors the turkeys, two adjacent turkeys will never end up on the same space since the distance between them is always odd.

Case 2: n is odd and composite.

We write $n = dk$ for some integers $2 < d, k < n$. The vertices of n form k distinct regular d -gons. Label the d -gons $0, \dots, k - 1$. For the i -th regular d -gon, we label its vertices $id + 1, \dots, id + d$. Look at two turkeys that are spaced d players apart. Each of the vertices of one regular d -gon are all either less than or greater than each of the vertices of a different d -gon. The two vertices clockwise of the two turkeys lie on the same d -gon while the two vertices counterclockwise of the two turkeys lie on the same d -gon that is distinct from the previous one. Therefore, no matter if Beto whistles or claps, the turkeys move in the same direction and always stay d apart. It is impossible to get these two turkeys to the same space.

Case 3: n is an odd prime.

We show that Beto may win. Note that whenever two turkeys go to the same spot, they'll follow the instructions as one, so Beto may as well eat one of them. We show that Beto may take any pair of turkeys on the board, move them onto the same spot, and eat one of them. After enough moves, only one turkey is left signifying that all turkeys are at the same spot.

For a pair of turkeys A, B , the distance between them traveling from A to B clockwise and from B to A clockwise have different parities since the two distances add up to n . Define the “effective distance” as the even value of the two.

Beto uses the following algorithm to take the effective distance to zero.

1. If either the effective distance decreases if Beto claps, the turkeys must have moved towards each other on the even-distanced side decreasing their effective distance by two. Beto will clap.
2. If either the effective distance increases or if the turkeys swap spaces if Beto claps, Beto will whistle. This is because the turkeys will then travel in a different direction meaning their effective distance decreases by two.
3. The only remaining case is if the effective distance does not change, and the turkeys do not swap spaces if Beto claps. In this case, the turkeys must move in the same clockwise or counterclockwise direction upon a clap meaning they’ll move counterclockwise or clockwise respectively upon a whistle. Beto will either whistle or clap to move both turkeys clockwise.

Note that (1) and (2) decrease the effective distance while (3) keeps it the same. Therefore, the only way the algorithm gets stuck in an infinite loop is if Beto endlessly plays (3). This would imply that, for some d that is the current effective distance between the turkeys, at each pair of vertices A, B that are d apart, the turkeys at A, B would move in the same direction upon either instruction. Therefore,

$$(\text{CW neighbor of } A > \text{CCW neighbor of } A) \text{ XOR } (\text{CW neighbor of } B > \text{CCW neighbor of } B) = 0$$

Let $a_1, a_2, \dots, a_n$ be the labels of the vertices in clockwise order. The above statement is equivalent to,

$$(a_i > a_{i+2}) \text{ XOR } (a_{i+d} > a_{i+d+2}) = 0$$

Where indices are taken modulo n . WLOG, let $a_1 > a_3$. We have,

$$a_1 > a_3 \Rightarrow a_{1+d} > a_{3+d} \Rightarrow a_{1+2d} > a_{3+2d} \Rightarrow \dots$$

Since n is prime and $1 \leq d < n$,

$$\{1, 1+d, 1+2d, \dots, 1+(n-1)d\} \equiv \{1, \dots, n\} \pmod{n}$$

Therefore, $a_i > a_{i+2}$ for all i .

$$a_1 > a_3 > a_5 > \dots > a_{2n+1} = a_1 (!)$$

52 clubs; 20.5 hours